

Full Length Article

Optimum sizing of hybrid renewable energy system with biomass backup of Egypt's Western Desert

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ABSTRACT

This paper presents an optimized design of an off-grid microgrid capable of fulfilling the year-round power demands of a remote area in Egypt. The proposed system combines photovoltaic, Wind Turbine (WT), backed by batteries, and biomass technologies to ensure consistency and eco-friendliness with fewer pollutants. A comprehensive analysis using HOMER Pro software was utilized to identify the optimum design, considering the energy demand, renewable resource availability, and net present cost (NPC) of the system. The results show that the optimal design includes a solar Photovoltaic (PV) system, WT, backed up by batteries, and a stand-by biomass generator (BIOMG). The design can significantly decrease reliance on conventional energy sources and promote the growth of renewable energy in the region. This study is considered the cheapest model in this configuration (PV, WT, Batteries, BIOMG) if compared to previous studies in the same location through a total NPC of USD3,483,519.00 and the best cost of energy (COE) of USD0.118. Additionally, the proposed system decreases emissions that contribute to global warming with a GHG of 28,516 tons/year. A sensitivity analysis of the uncertainty of renewable resources and economics has been conducted. The sensitivity analysis illustrates the variety of results.

1. Introduction

Electric energy is considered a furthermost significant pointers of environmental, financial, and industrial development in nations. For decades, the need for energy is growing as a result of population growth, civilization, plus industrialization, which has led to an increase in welfare. Despite using of conventional energy sources worldwide for many decades, there has recently been a substantial increase in using of different renewable energy sources (RES), for instance, solar, wind, hydroelectric, green oxygen, and biomass. This can be attributed to their environmentally friendly, sustainable, and clean characteristics. The release of carbon dioxide (CO₂) and other contaminants into the atmosphere through environmentally unfriendly manner, stemming from traditional energy sources predominantly fueled by fossil fuels, contributes to the emission of greenhouse gases (GHGs), exacerbating environmental shifts and the rise in global temperatures. Consequently, reliance on the RES is diminishing globally [1,2].

In several countries, traditional fossil fuel-based power plants (such

as coal, gas, and oil) produce around 70 % of electricity for different end-use applications, including residential, agricultural, industrial, and so on [3]. The rise in fossil fuel consumption as well as extreme carbon emissions is harmful to the environment, prompting policymakers to seek for other solutions to combat global energy demands and climate change [4]. Many countries have chosen to limit carbon emissions that are causing climate change and global warming as per the Paris Agreement made at the United Nations Climate Change Conference 2015. GHG emissions should be reduced by 40 % by 2030 and by 80 % by 2050. To achieve this goal, it is essential to substitute conventional energy sources with renewable ones. By 2050, renewable energy sources are predicted to contribute 30 % of the energy supply. It is assumed that renewable energy sources will provide all the energy needed. Consequently, the significance and application of RESs are increasing globally. This issue is receiving particularly high attention from power generation companies, energy policymakers, and governments [5–10].

Solar and wind constitute two primary RESs, comprising a significant share of the global energy market [11]. However, RESs' applications are restricted due to instability caused by uncertain and intermittent

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Nomenclature**PV SYSTEM**

$T_{cell}(t)$	Cell temperature at time t
T_r	Standard cell temperature
T_{NO}	Normal cell temperature
N_{PV}	number of PV modules
η_{wv}	Wiring efficiency
η_{pv}	System efficiency
P_{rated}^{PV}	PV –rated power
$R_{int}(t)$	Solar radiation
R_{nom}	solar radiation considering the standard condition
γ_T	The temperature coefficient of the PV.

Wind Turbine

S_1	Wind speed at height H_1 .
S_2	Turbine hub with height H_2 .
β_{wt}	WT friction coefficient.
P_{rated}^{wt}	Rated power from WT at rated wind speed
P_{wt}	Generated power from WT.
S_{cut}^{in}	Cut –in wind speed.
S_{cut}^{off}	Cut –off wind speed.
$S(t)$	Wind speed at time t .
N_{wt}	No. of wind turbines in the wind farm.
η_{wt}	Wind turbine efficiency.

Biomass System

$P_{BG}(t)$	Biomass output power
$F_{ConsBG}(t)$	Average fuel consumption / hour
η_g	Generator efficiency
LHV_{pg}	Lower heat value of the product gas
m_{pg}	Mass flow of product gas

m_b	Mass flow of biomass
LHV_B	Lower heat value of biomass
N_{BG}	Number of generators
B_{rated}	consumption rate / hour (kg/h)
F_m	Marginal fuel consumption
F_0	no –load fuel consumption
PG_{rated}	Rated power

Battery System

$SoC_{Bat}(t)$	State of charge
SoC_{max}	Maximum state of charge
SoC_{min}	Minimum state of charge
B_{CH}	Charging energy
B_{DIS}	Discharging energy
$P_{re}(t)$	Hourly cumulative power output from the PV, wind turbines, and biomass systems
η_{CH}	Battery charging efficiency
η_{DIS}	Battery discharging efficiency
σ	Self –Discharge Rate
η_{inv}	Inverter efficiency
$P_L(t)$	Hourly load demand

Cost Analysis

C_{ann}^{tot}	Total annual energy cost
EC	Average cost of useful energy (\$/kWh).
C_{ann}^{cap}	Annual interest on the capital cost.
C_{ann}^{fuel}	yearly fuel cost of the biomass feedstock.
$C_{O\&M}$	Operation and maintenance cost.
$C_{rep,i}$	Replacement cost of each subsystem
CRF	Capital recovery factor.
P_L^{tot}	The total demand accumulated over 8,760 hours.

weather conditions [12]. Energy security, resilience, and sustainability are acknowledged to be effectively addressed by micro-grid systems that combine conventional and RESs with energy storage units, particularly in isolated, rural, or island locations where grid connections are costly and complicated [13].

Several recent studies have examined the efficacy of various optimization strategies for determining the finest configuration of Hybrid Energy Systems (HESs). Diab et al. [14] suggested a hybrid off-grid RES system that combine PV and WT components incorporating a diesel generator (DG) and batteries units for an isolated location in Egypt. Therefore, four techniques—Whale Optimization Algorithm (WOA), and other algorithms—were applied and evaluated in this study. Among these methods, the WOA strategy outperformed the others for detecting the furthestmost appropriate configuration of the HES, intending to minimize energy costs.

In another study by Ramli et al. [14], an applied optimization technique called the Multi-Objective Self-Adaptive Differential Evolution (MOSAde) approach is employed to determine the best configuration of an analyzed RES. Therefore, three various arrangements of hybrid RESs containing PV and WT, battery storage, and DG systems were considered. The primary goals of the study were to reduce the COE provided by RESs as well as minimize the loss of power supply probability (LPSP).

The Grasshopper Optimization Algorithm (GOA) has been employed for getting the finest rating of off-grid Hybrid Renewable Energy Systems (HRES) consisting of five housing units in Yobe state, Nigeria [15]. This system comprised WT, PV, DG, and battery units. To validate GOA results, they have been associated with those of other optimization methods. However, the time required for implementation using the presented method was not specified.

In a study conducted for Abu-Minqar, an optimization model was proposed for diminishing NPC, and COE, while also improving the reliability of the power supply. The consequences of the applied technique provided an optimum arrangement of HRES reflecting the NPC, COE, and LPSP [16].

One of the most effective methods for ensuring that the components of energy systems are planned optimally is by utilizing HOMER which has been created by the NREL [17]. A wide variety of systems, such as WT, PV, converters, diesel generators, BES, etc., are included in this software. HOMER reduces the NPC for addressing the optimal planning issues. A variety of outcomes are displayed by HOMER software, including cash flows, sensitivity analysis, ideal sizes, and other economic and technical analysis [17]. It has a wide range of capabilities for analysis and optimization. The facility to design the AC and DC power systems is one of HOMER-Pro's key characteristics, making it particularly useful for examining hybrid systems that use both power sources [18].

A comparison between several tools that can reach the best sizing of HRES has been introduced in [19]. The comparison included many softwares, such as (HOMER, IHOGA, Hybrid 2, RETScreen, and TRNSYS) and traditional methods, such as the analytical Method, Probabilistic method, Iterative approach, and artificial intelligence method. Conventional approaches excel in optimizing size fields more effectively than commercial tools, offering rapid outcomes and the ability to tackle multi-objective challenges. The furthestmost significant problem of this approach was its complexity as well as lack of versatility. On the other hand, HOMER optimization is simple and can be used for designing and sizing any hybrid system [19].

S. Mandal [16] studied and evaluated six possible stand-alone HRES configurations incorporating WT, PV, battery units, and DG in a remote

rural Bangladeshi location. The criteria of lowest energy cost and least amount of greenhouse gas emissions was used to specify the best configuration. The HOMER program is the optimal method employed in this investigation.

The development of an efficient HRES system to address these issues has been made possible by the introduction of various models and optimization methods by several researchers. Table 1 presents some approaches to developing hybrid optimization models, and most of these studies did not study greenhouse gas emissions, which is considered a gap in these studies because each CO₂ emission affects global warming.

Despite the recent advances in HRESs, there is still a need for further investigation, especially in the context of developing countries. A major research gap is the lack of studies focusing on the optimal design of HRESs for remote rural areas, considering the specific local conditions and resources. So, this paper is motivated by the need to address this research gap and provide valuable information for policymakers, researchers, and industry practitioners on the application of HRESs in remote rural areas. In addition, the paper addresses environmental concerns by considering CO₂ and GHG emissions, filling the identified research gap of their omission in Table 2.

This paper will make the following contributions:

1. It presents an optimal design for a HRES with a backup biomass generator in the remote area of Abu-Minqar in Egypt.
2. It demonstrates the viability and efficiency of HRESs in isolated rural areas.
3. It provides valuable information for policymakers, researchers, and industry practitioners on the operation of HRESs in remote rural areas.
4. Specifically, the paper presents a proposed HRES that can meet all electrical loads in Abu-Minqar with the lowest NPC and leveled cost of energy (LCOE) compared with other studies considering the same area. This is due to the use of the biomass generator as a backup source for only 505 h/ year which reduces the Pollutive emissions

Table 2

Hybrid System Parameters.

Parameter	Value	Parameter	Value
PV System		Inverter Device	
PV Panel cost (\$/kW)	24.6	η_{inv} Inverter efficiency (%)	95
γ_T (%)	0.0045	Max. Power (kW)	487
η_{pv}	13 %	Inverter lifetime (Year)	10
T_r (°C)	45.7	Inverter cost (\$/kW)	711
R_{Nom} (W/m ²)	1000	Inverter replacement cost (\$/kW)	650
Rated Power (kW)	1	Wind Turbine System	
Length (mm)	1625	Model	Pitchwind –PIT30
Width (mm)	1019	Rated Power (kW)	30
Thickness (mm)	46	Height (m)	50
Lifetime	25	η_{wt} (%)	80
PV replacement cost	23	$S_{cut}^n, S_{rated}, S_{cut}^{off}$ (m/s)	[2.5,12,25]
Battery System		Lifetime (Year)	20
Model	Surrette 6CS 25P	Cost (\$/kW)	3200
Nominal battery voltage (V)	6	Biomass Generator System (Auto sized)	
Nominal battery capacity (Ah)	820	LHV _B ((MJ/Kg)	4.766
Storage capacity of battery (kWh)	6.9	LHV _{pg} (MJ/Kg)	0.01
η_{CH} (%)	80	η_g (%)	80
η_{DIS} (%)	80	PG _{rated} (kW)	460
σ	0.005	F0, Fm (kg/hr./50 kW)	0.0644, 0.2998
No. of strings	39	BGcost (\$/kW)	4000
Battery cost (\$/unit)	300	SBG (Year)	25

and meets COP21 objectives. The greenhouse gas emission is also studied.

5. This research enhances the expanding expertise on HRESs and provides valuable information for the implementation of such systems in

Table 1

A sampling of the current research of HRESs.

Ref.	Year	Grid	Site	Components								Approaches	Research Gap
				PV/ CPVT	Wind	FC	Biomass	Hydrogen storage	Electrolyzer	BG/ DG/ MTG	BESS		
[20]	2019	×	Iran	✓	✓	✓	×	×	×	×	×	Flower pollination algorithm	Environmental consequences are not taken into account.
[21]	2020	×	Qatar	✓	✓	×	✓	✓	×	×	✓	PVsyst, HOMER and EES	The thermal loss generated by CPV/T is not considered.
[22]	2022	×	Egypt	✓	×	✓	✓	×	×	×	×	Mayfly optimization algorithm	Emissions of nitrogen oxides from biomass are not considered in the GHG calculations
[23]	2022	×	Iran	✓	✓	×	×	×	×	✓	✓	HOMER	lowest LCOE only has 37.1 % RF, but it still produces a lot of CO ₂ emissions.
[24]	2023	×	Cameroon	✓	×	×	×	×	✓	✓	✓	HOMER Pro	DG's environmental impact is not involved
[25]	2023	×	Afghanistan	✓	✓	×	×	×	×	✓	✓	HOMER Pro, simplex algorithm	The effects of CO ₂ emissions and GHG are not being observed.
[26]	2023	×	Egypt	✓	✓	×	×	×	×	✓	✓	HOMER Pro	Although the system has studied CO ₂ emissions, GHG are not studied.
[27]	2023	×	Egypt	✓	×	×	×	×	×	✓	×	MATLAB and HOMER Pro	Emissions of nitrogen oxides from biomass are not considered in the GHG calculations. Although the system has excess load, there are unmet loads and capacity shortage/ year.

remote rural areas. The paper addresses a research gap by exploring the optimal design of a HRES using the HOMER Pro software. This software, while powerful for such designs, is not commonly utilized in developing countries due to its expense. However, the study demonstrates that even with limited resources, HOMER Pro can be effectively employed to design and optimize HRESs in these regions.

The remaining sections of the paper are categorized into three main ones. The first section, System Description, provides a detailed overview of the HRES's components, cost analysis, optimization objectives, and operational strategy. The second section, case study and results presents the outcomes of the optimization process, including component sizes, costs, system performance metrics, and empirical evidence of feasibility. It demonstrates the viability besides the performance of the planned configuration, providing valuable insights for researchers, policy-makers, and industry practitioners. The final section, "Conclusions," presents a concise summary of the study's principal findings and their ramifications for the configuration and deployment of HRESs.

2. System design

The study in this paper investigates and develops the optimum sizing and energy management strategy for a HES that integrates Photovoltaic (PV) units, Wind Turbines (WT), and BIOMG as the primary power source, with batteries used for energy storage. Dump loads are a critical component of off-grid hybrid systems, as they help to dissipate excess power in case of fully charged batteries or when there is an oversupply of power that is not currently needed. The system is designed to provide energy to a rural village of Abu-Minqar, located in Egypt. The planned configuration is shown in Fig. 1, which has been optimized using HOMER platform as a simulation tool to obtain the optimum arrangement and size of the HES.

2.1. Photovoltaic system

The performance of a solar module has been exaggerated by variations in the weather and environmental conditions at the site of installation. The PV output power (P_{PV}) may be mathematically represented as [14,23,28,29]:

$$T_{cell}(t) = R_{int}(t) \frac{T_{NO} - 20}{0.8} + T_{amb} \quad (1)$$

$$P_{PV}(T) = N_{PV} P_{rated}^{PV} \eta_{pv} \left(\frac{R_{int}(t)}{R_{nom}} \right) (1 - \gamma_T (T_{cell}(t) - T_r)) \quad (2)$$

Solar irradiance data have been generated in Homer software database over 20 years for the proposed site of installation (Abu-Minqar).

2.2. Wind turbine system

The WT comprises three fundamental components: tower, blades, and generator, which can be used for converting the kinetic energy from the wind into energy [30,31]. The generated energy by WTs may be influenced by wind speed and blades' design [32]. As wind speed generally increases with elevation, it is crucial to calculate the wind speed at a given height, H_2 , using the following equation [33].

$$\frac{S_2}{S_1} = \left(\frac{H_2}{H_1} \right)^{\beta_{wt}} \quad (3)$$

Along with the International Electro-technical Committee (IEC) [29,34], the friction coefficient values under different wind conditions are established. In nominal wind conditions, the friction coefficient is typically 0.20, while under more intense wind conditions, it decreases to 0.11. The production of the WTs (P_{wt}) has been presented using the following formula [35,36].

$$P_{wt} = \begin{cases} 0 & S(t) < S_{cut}^{in} \text{ or } S(t) > S_{cut}^{off} \\ N_{wt} P_{rated}^{wt} \eta_{wt} \left(\frac{S^2(t) - S_{cut}^{2}}{S_{rated}^2 - S_{cut}^2} \right) & S_{cut}^{in} < S(t) < S_{rated} \\ N_{wt} P_{rated}^{wt} \eta_{wt} & S_{rated} < S(t) < S_{cut}^{off} \end{cases} \quad (4)$$

2.3. Biomass system

This paper employs a fixed-bed gasifier, utilizing the gasification process, and a gas turbine to process Corn Stover (CS), a substantial agricultural waste frequently underutilized in Egypt. The units use CS as biomass feedstock in a gasifier, breaking biomass particles at over 700 °C in four stages. The particles are recombined with air as a gasification agent, forming synthetic gas for electricity production in turbines [37,38].

The efficiency of the syngas generated through the gasification development η_g , generated power $P_{BG}(t)$, and normal fuel consumption/

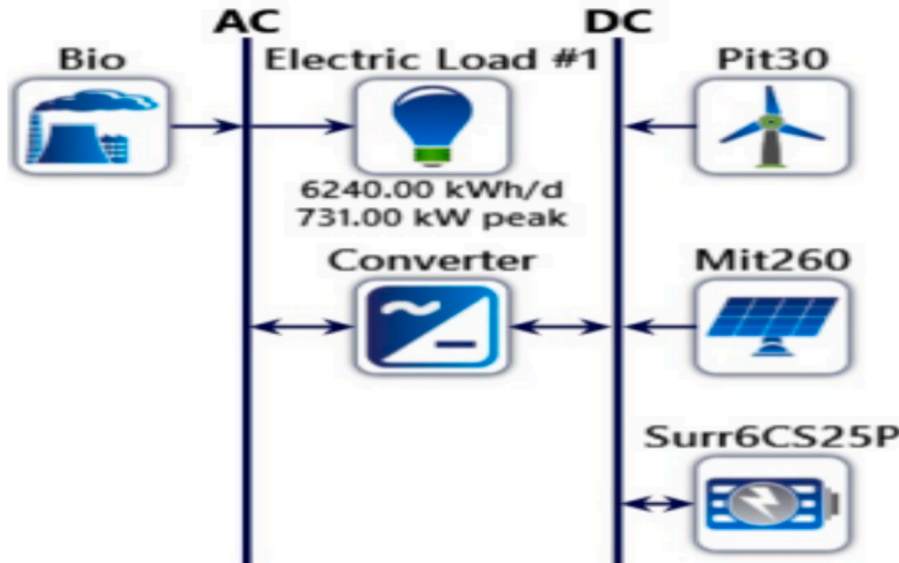


Fig. 1. Proposed configuration of a studied HRES.

h $F_{\text{ConsBG}}(t)$ can be planned as, [23,37];

$$\eta_g = \frac{LHV_{pg}m_{pg}}{LHV_Bm_B} \quad (5)$$

$$P_{BG}(t) = \frac{N_{BG}}{F_m} \left(\frac{\eta_g LHV_B B_{\text{rated}}(t)}{LHV_{pg}} - F_0 P_{G_{\text{rated}}} \right) \quad (6)$$

$$F_{\text{ConsBG}}(t) = \frac{LHV_{pg}}{\eta_g LHV_B} (N_{BG} F_0 P_{G_{\text{rated}}} + F_m P_{BG}(t)) \quad (7)$$

$$\text{GHG} = \text{Energy Consumption} * \text{Emission Factor} * \text{GWP} \quad (8)$$

$$\text{Total Biomass Energy (kJ)} = \text{Total Consumption} * LHV_B \quad (9)$$

$$\text{Energy Consumption (kWh)} = \text{Total Biomass Energy} * 0.00022778 \quad (10)$$

Equations (8)–(10) provide a description of the technique used to estimate the air emissions of GHG from agricultural waste. The property of gas affects the degree to which GHGs/unit emitted impact global warming. To assess the emissions of various GHGs, CO₂ and nitrogen oxides are used to categorize all emissions. The CO₂ and nitrogen oxide reference points are used to measure these emissions and have a GWP of 1. CO₂ has a GWP that is greater than that of all other GHGs listed in the Kyoto Protocol [23,39].

2.4. Battery system

Battery storage is crucial when PV units, WTs, and biomass systems exceed load demand. These batteries discharge when energy from sustainable sources fails for fulfilling demand or for system maintenance. Among various battery types, the lead-acid battery has been employed [29]. The charge status ($SoC_{\text{Bat}}(t)$) serves as an indicator of the battery capacity. The maximum (SoC_{max}) and minimum (SoC_{min}) are considered the charging process limits. The SoC_{Bat} , the charging and discharging energies (B_{CH}) and (B_{DIS}) are presented as, [14,40]:

For charging process:

$$B_{\text{CH}}(t) = \left(P_{\text{re}}(t) - \frac{P_{\text{re}}(t)}{\eta_{\text{inv}}} \right) * \Delta t * \eta_{\text{CH}} \quad (11)$$

$$P_{\text{re}} = P_{\text{pv}} + P_{\text{wt}} + \frac{P_{\text{BG}}}{\eta_{\text{inv}}} \quad (12)$$

$$SoC_{\text{Bat}}(t) = SoC_{\text{Bat}}(t-1) * (1 - \sigma) + B_{\text{CH}}(t) \quad (13)$$

For the discharging process:

$$B_{\text{DIS}}(t) = \left(\left(\frac{P_L(t)}{\eta_{\text{inv}}} \right) - P_{\text{re}}(t) \right) * \Delta t * \eta_{\text{DIS}} \quad (14)$$

$$SoC_{\text{Bat}}(t) = SoC_{\text{Bat}}(t-1) * (1 - \sigma) - B_{\text{DIS}}(t) \quad (15)$$

2.5. Cost analysis

The total annual cost ($C_{\text{ann}}^{\text{tot}}$), COE, and NPC have to be calculated somewhere the primary aim is to adjust the values of the HRES fixed capacities to assure a reliable supply and low cost. The COE can be calculated as [41,42]:

$$C_{\text{ann}}^{\text{tot}} = C_{\text{ann}}^{\text{cap}} + C_{\text{ann}}^{\text{fuel}} + \sum_{m=1}^S C_{O\&M,i} + \sum_{m=1}^S C_{\text{rep},i} \quad (16)$$

$$\text{CRF} = r(r+1)^S / (r+1)^S - 1 \quad (17)$$

$$\text{EC} = \frac{C_{\text{ann}}^{\text{tot}}}{P_L^{\text{tot}}} \quad (18)$$

P_L^{tot} constitutes the aggregate of the entire load demand over 8760 h.

3. Simulation tool

3.1. HOMER software

HOMER serves three primary functions: simulation, optimization, and sensitivity analysis. To assess the feasibility and life-cycle cost of a certain microgrid, HOMER performs hourly simulations throughout the entire year. By simulating various system configurations, HOMER identifies the setup that meets the technical requirements while minimizing life-cycle costs.

In order to assess the impact of uncertainty or variations in model inputs, HOMER conducts multiple optimizations under different input scenarios. The optimum setting for a variable is one that the designer of the system has control over, for instance the combination of system components and the size or number of each. The impact of uncertainties in variables beyond the designer's control, such as average wind speed or future fuel prices, is assessed through sensitivity analysis. The connection among simulation, optimization, and sensitivity analysis utilizing HOMER is depicted in Fig. 2. The simulation oval is enclosed by the optimization oval to represent the fact that different simulations are combined into a single optimization. Similar to how the optimization oval includes the sensitivity analysis oval, a sensitivity analysis is made up of many optimizations, considering WT cost, the price of biomass fuel and battery, and other factors that affect sensitivity [43].

3.2. Operating strategy

The prime sources can be considered as PV and WT, with batteries ensuring continuous power supply and improving the supply to loads. Moreover, the management approach of the HRES to fulfill the load demand relies on the Pre and the inverter power P_{inv} . [29]. The biomass generator is considered as a backup source for more reliability and to have reliable system, the generator is work when the battery charge is below 40 %.

$$P_{\text{inv}}(t) = \frac{P_L(t)}{\eta_{\text{inv}}} \quad (19)$$

Fig. 3 illustrates the flow chart of the applied operating strategy which can be summarized as follows:

1. When P_{re} meets the demand ($P_{\text{re}}(t) = P_{\text{inv}}(t)$), the $SoC(t)$ of the battery bank remains constant, and $LPS(t) = 0$.
2. When P_{re} produced from the system exceeds the demand ($P_{\text{re}}(t) > P_{\text{inv}}(t)$), as well as the $SoC(t) < SoC_{\text{max}}$, The extra power ($PS(t)$) will charge the battery to the maximum limit SoC_{max} .
3. If $SoC(t) = SoC_{\text{max}}$, SoC remains constant as SoC of previous preceding hour ($SoC_{\text{Bat}}(t) = SoC_{\text{Bat}}(t-1)$), Simultaneously, the extra generation is considered as a dummy load (ED-load).
4. If the demand is larger than the RES generated power, as well as the SoC exceeds the SoC_{min} ($P_{\text{re}}(t) < P_{\text{inv}}(t)$ and $SoC_{\text{Bat}}(t-1) * (1 - \sigma) > SoC_{\text{min}}$), the stored energy can be utilized to fulfill the demand.
5. If $SoC_{\text{Bat}}(t) < 40\%$, then the backup biomass generator shall start to feed the loads.

4. Results discussion

4.1. System parameters and settings

This paper examines the real case of Abu-Minqar, a rural village in the Farafra Oasis, Egypt. The selected location for installing the HRES is at 26.0°30.30' N, 27.0°39.70' E. Fig. 4 shows the geographic location in this Rudy area.

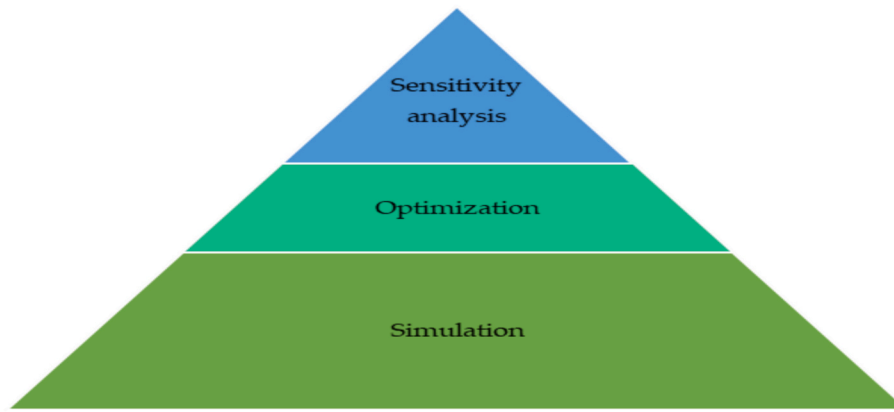


Fig. 2. Relation between Simulations, Optimization, and sensitivity analysis in HOMER.

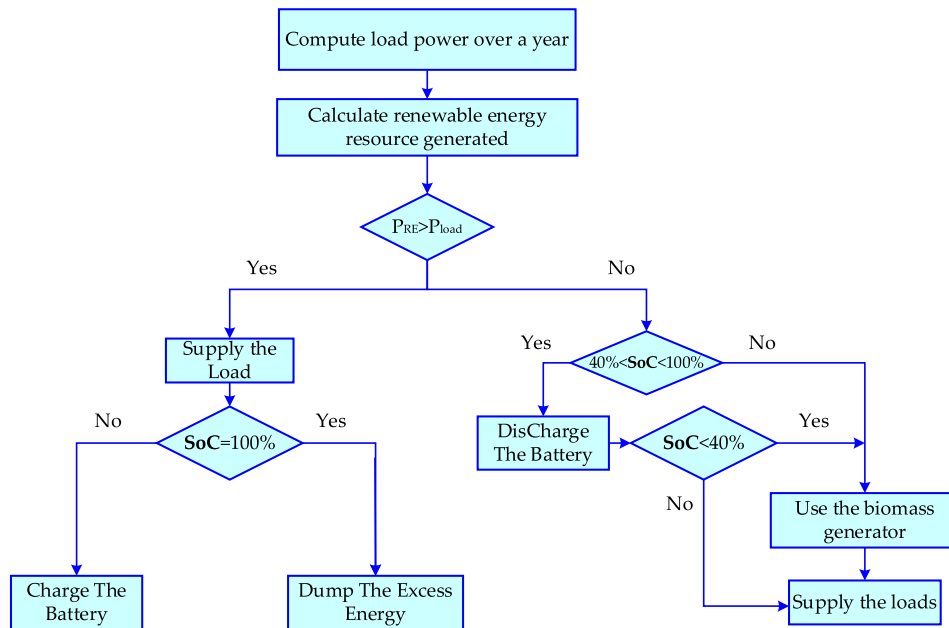


Fig. 3. Homer flow chart of the system.

The village is renowned for its profuse making of crops, agricultural products, and livestock, constituting a significant source of biomass. Agriculturalists predominantly encourage cultivating field-grown crops, including maize, clover, beans, and wheat, utilizing flood watering techniques. Corn is recommended as the preferred biomass for the Gas conversion process because of its widespread accessibility. The month-by-month usage frequency throughout the year is illustrated in Fig. 5.

Figs. 6 and 7 display the solar global horizontal irradiation (GHI) and the monthly wind speed averages, compiled for two decades for Abu-Minqar, sourced from the NASA surface meteorology.

In this area, the basic loads are represented by household and public utilities load with a maximum load of 410 kW and an average load of 260 kW. The hourly load curve during the year is shown in Fig. 8 and a sample from 3000 h to 3500 h is given in the same figure. The monthly average load per day is shown in Fig. 9 and the yearly demand profile has been given in Fig. 10.

To simulate the energy and heat co-generation system and optimize the rating of each component, it is required certain technical specifications and the costs of the associated equipment. Consequently, Table 2 shows all system parameters as per the data input for the simulation purpose.

4.2. Results

HOMER Pro software has been used for optimization the HRES. Table 3 shows the rating of each component and it is clear that the PV and Biomass generator can be loaded with full loads while the WT can partially be loaded. The battery system is the backup for the wind and the PV.

The final NPC of the system is \$3,483,519.00 with a LCOE of \$0.118. The NPC value for each component is shown in Table 4 while the chart shows, in Fig. 11 the NPC percentage for each component.

The total annual cost is 269,465.00 \$ while the breakdown of all component's annual cost is shown in Table 5.

The proposed system is a reliable system as there are no capacity shortage or unmet loads while there is a yearly excess load with 804,156.00 kWh/year while the total yearly generated energy is 3,486,265.00 kWh/year with only 23 % of the total generated energy. Table 6 shows the production summary for each component.

From Table 6, it is clear that the PV has the major participation by 95.2 %. Table 7 shows the operating hours for each component.

The system is reliable, and it is clearly shown that the operating hours of components can cover all operating hours for loads (8760 h/Year) and can cover all loads capacities while the system net present cost

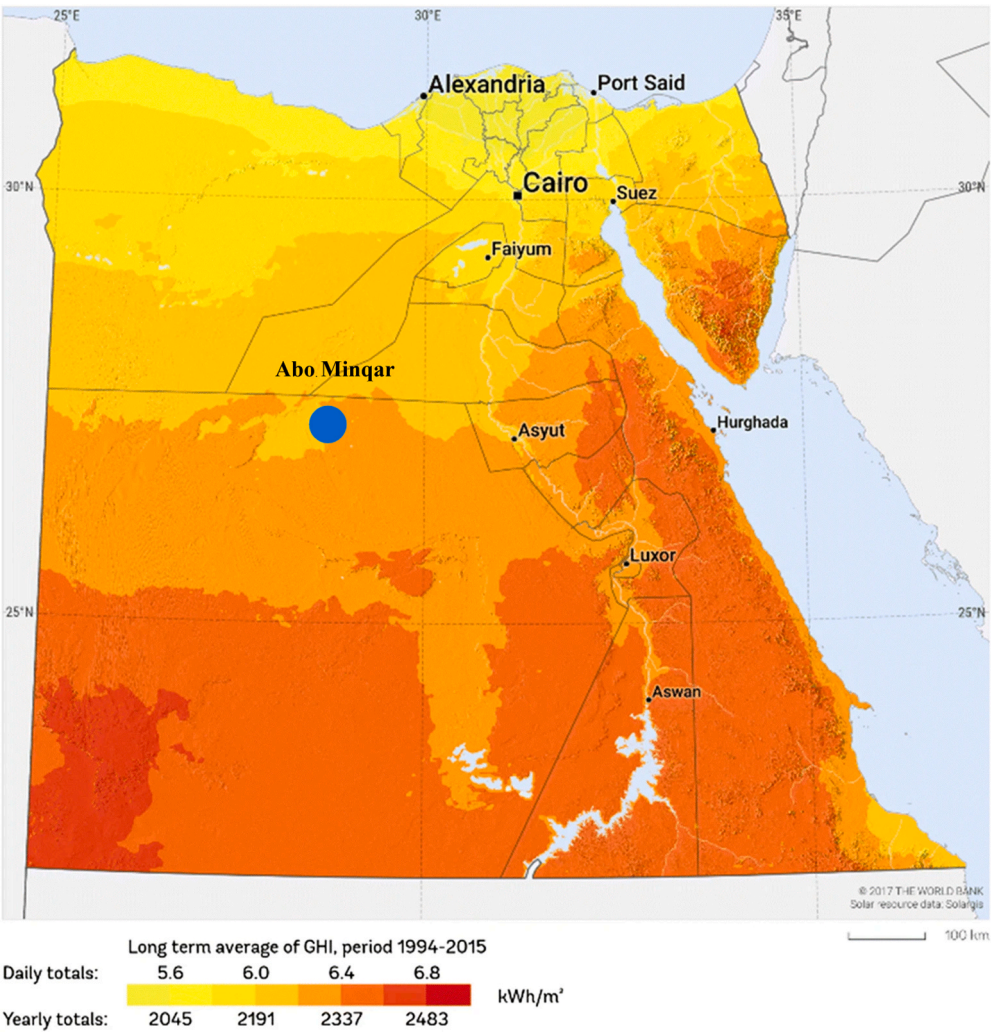


Fig. 4. Case Study Location (Abu–Minqar Location).

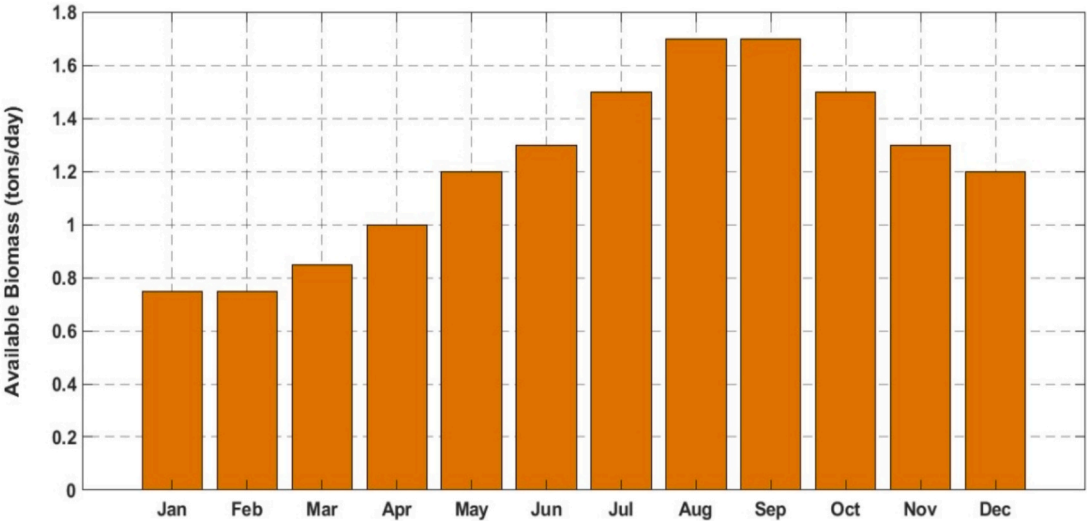


Fig. 5. Yearly Biomass Consumption Rates.

is optimized also, it is clear that the biomass generator is fully back-up to the system converter with 214 starts/Year.

With higher living standards, more economic activity, safety, job

possibilities, and Green growth HRES can benefit rural communities. The hybrid power system is also significantly increasing its contribution to a greener environment. Another advantage of selling the excess

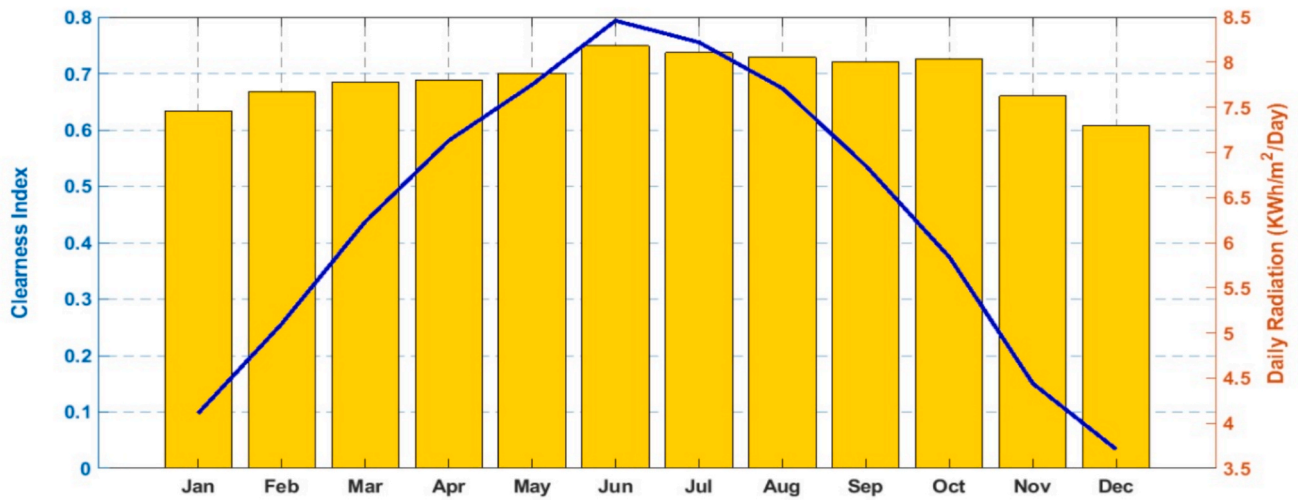


Fig. 6. Yearly Solar irradiation.

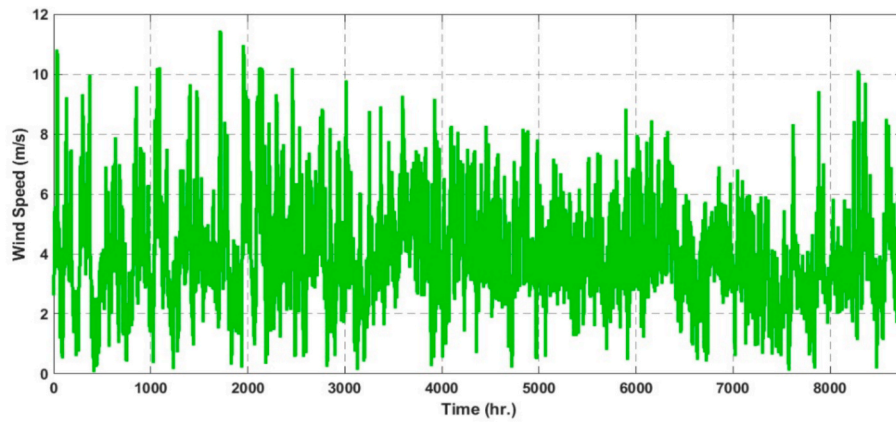


Fig. 7. Yearly Wind Speed Data.

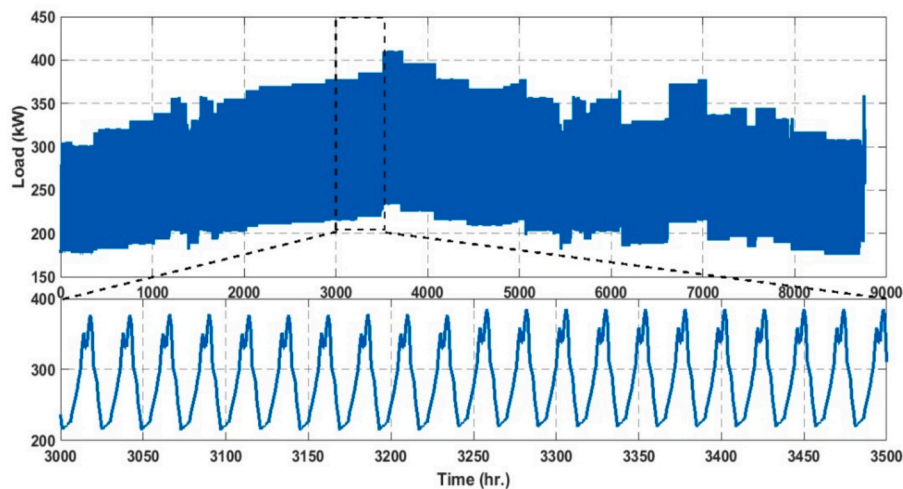


Fig. 8. Yearly Load Profile (8760 h) and Sample Load Profile from 3000 hr. to 3500 hr.

energy to regional customers so the national grid would significantly diminish it.

In order to deploy the standalone hybrid power system, government and business organizations need to take an initiative in order to remove any obstacles. Incorporating renewable energy sources into the current

system reduces greenhouse gas emissions and enhances the reliability of systems, ensuring the ability to fulfill peak customer's needs. This study highlights a significant decrease in the fuel utilized by the designed biomass generator, which in turn reduces GHG emissions. In this analysis, biomass is taken into account as fuel. The amount to an average

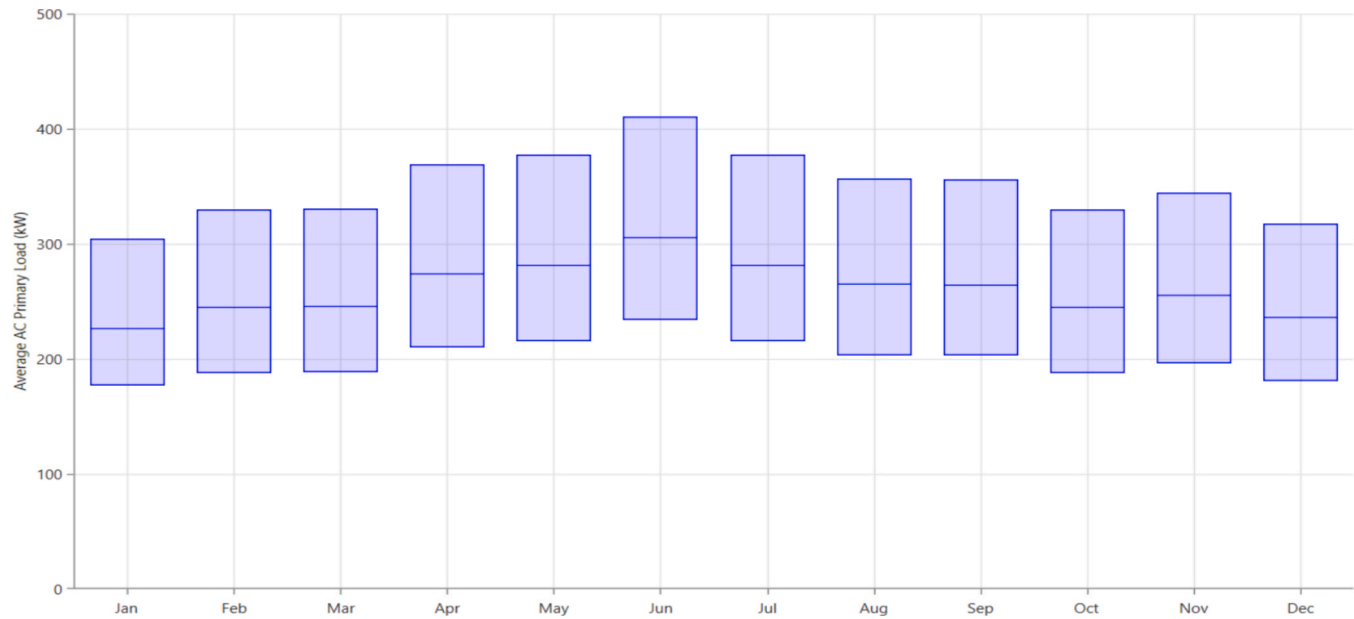


Fig. 9. Daily Profile for 24 h.

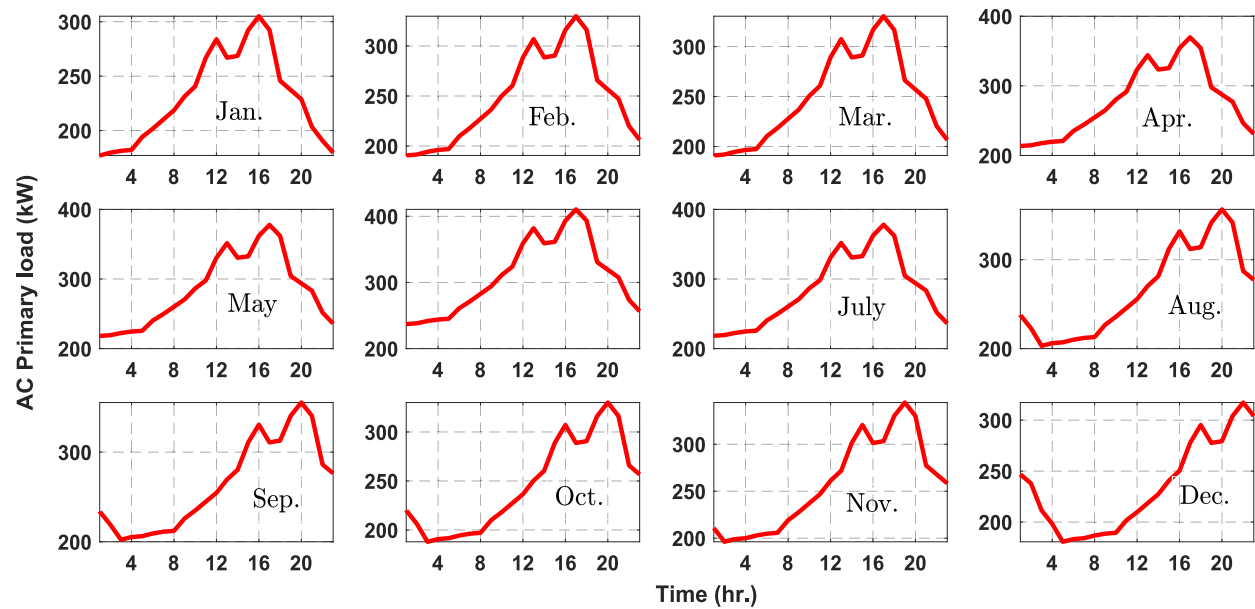


Fig. 10. Yearly Load Profile.

Table 3
Ratings for each component.

Component	Name	Size	Unit
Biomass Generator	Autosized by HOMER	460	kW
PV	Mitsubishi Electric260PV-MLU260HC	1,749	kW
Storage (total rating is 2,487 kWh)	Surrette 6 CS 25P (nominal capacity 6.91 kWh)	39	Strings (Each string is 40 Battery)
Wind Turbine	Pitchwind 14 m 30KW	30	kW
System Converter	System Converter (Inverter)	400	kW

feedstock/day is 109 kg. The yearly consumption of feedstock is 39.9 Ton. Table 8 shows the yearly emissions of the biomass generator which are minimized.

Table 4
NPC for Each Component.

Name	Total Cost (\$)
Biomass Generator	1,960,00.00
Mitsubishi Electric260PV-MLU260HC	172,185.00
Surrette 6 CS 25P (nominal capacity 6.91 kWh)	881,314.00
Pitchwind 14 m 30KW	109,357.00
System Converter (Inverter)	363,543.00
Total NPC	3,483,519.00

As a result, the ideal hybrid resources combination configuration for Abu-Minqar has been suggested. With this setup, the research area's essential energy consumption is met at the lowest possible cost. The shearing of PV is at its highest in every month's energy production. It is clear considering the maximum ratio of the demand, which is met

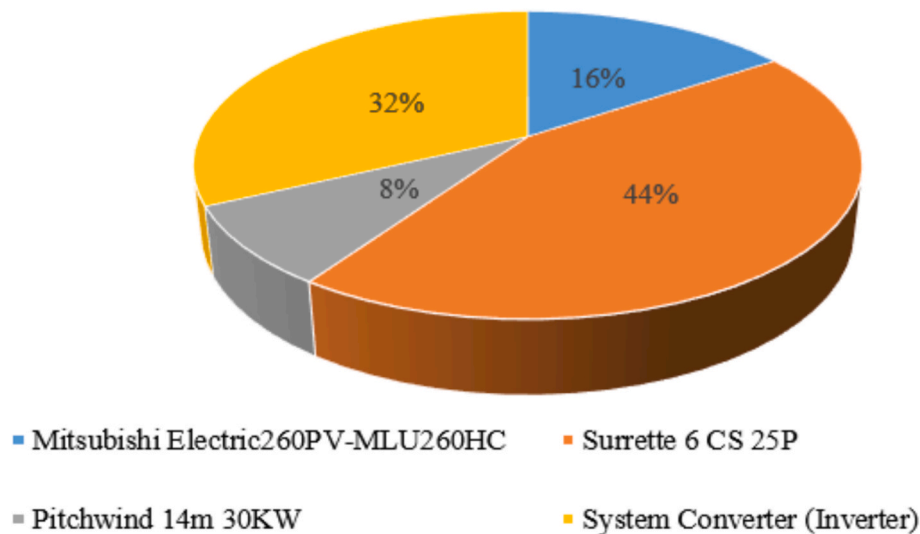


Fig. 11. NPC percentage for each component.

Table 5

Total Annual Cost for Each Component.

Name	Total Cost (\$)
Biomass Generator	151,392.00
Mitsubishi Electric260PV-MLU260HC	13,319.00
Surrrette 6 CS 25P	68,174.00
Pitchwind 14 m 30 kW	8,459.00
System Converter (Inverter)	28,122.00
Total annualized cost	269,465.00

Table 6

Yearly Production Summary.

Name	Production (kWh/year)
Biomass Generator	93,808.00
Mitsubishi Electric260PV-MLU260HC	3,318,818.00
Pitchwind 14 m 30 kW	73,639.00
Total Production	3,486,265.00

Table 7

Yearly Operating Hours.

Name	Hours
Biomass Generator (214 starts)	505
Mitsubishi Electric260PV-MLU260HC	4,378
Surrrette 6 CS 25P (autonomy)	24.8
Pitchwind 14 m 30 kW	7,888
System Converter (Inverter)	8,721

through solar energy, in the absence of solar energy, wind, and the battery bank has been employed for fulfilling the demand. However, the BIOMG is familiarized to supply the demand if neither solar PV, WT nor a battery bank can. The suggested hybrid system released only 6.27 kg Co₂ / year. The GHG is equal to 28,516 tons/year. This is superior to the GHG calculated in [22] (792.534 tons/year) that was investigated at the same location with the same loading profile.

Table 8

The Yearly Emissions of The Biomass Generator.

Pollutant	CO ₂	Carbon Monoxide	Unburned Hydrocarbon	Particulate Matter	Sulfur Dioxide	Nitrogen oxides
Quantity (kg/year)	6.27	0.658	0.287	0.00399	0	0.618

4.3. Sensitivity and uncertainty analysis

The sensitivity analysis has been conducted for more evaluating the studied HRES. Table 9 shows the sensitivity variables including wind, solar radiation and temperature combination also Biomass fuel scaled average, interest rates and discount rates. If a specific belief is met, all possible system combinations has been evaluated. In this regard 2 simulation times have been implemented to show the effect of the economics fluctuations and uncertainty, and to show the effect of resources uncertainty and economics uncertainty and to reach to the related solutions in each case.

4.3.1. Renewable resources sensitivity analysis

Several types of data, including load profile, component specifications, and meteorological data have been used in the simulation to consider the renewable resources sensitivity analysis. After that, it is well known that the data including uncertainty and every year is different from the previous one. The sensitivity analysis for the renewable resources is done considering that the wind speed is between 2 m/s – 10 m/s where the solar radiation is between 4 Kw/m²/d – 8 Kw/m²/d and the temperature rise is between 22.5 °C – 28 °C. The sensitive amounts of NPC, COE is shown in Fig. 11 with reaching to the maximum values of NPC and COE when the solar radiation reach to the maximum value (8 Kw/m²/d) and vice versa. The values are under the worst sensitive case of the temperature 28 °C while having minor changes from

Table 9

Sensitivity Variables.

Sensitivity Variables	Range
Case1: Renewable resources sensitivity analysis	
Wind speed	2 m/s – 10 m/s
Solar radiation	4 Kw/m ² /d – 8 Kw/m ² /d
Temperature rise	22.5 °C – 28 °C
Case2: Economic uncertainty sensitivity analysis	
Biomass fuel scaled average	1 t/d – 1.5 t/d
Expected inflation rate	2 % – 12 %
Discount rate	4 % – 25 %

if the temperature rise is 22.5 °C.

4.3.2. Economic uncertainty consideration

Forecasting the future value of cash in underdeveloped nations may be challenging due to significant fluctuations in the discount rate and inflation. Therefore, to achieve the most probable outcomes for the system's economics, both the minimum and maximum anticipated rates were included in the estimates. The nominal discount rate and inflation were assessed to fluctuate between 15–20 % and 10–25 %, respectively. The results in the surface plot (Fig. 12) proved that, if the discount rate is high and the inflation rate is low, then NPC, COE would be in lower value and vice versa. For example, when inflation = 2 % and discount rate = 25 %, the amount of NPC would equate to 2,783,427.00 \$. While the COE would be 0.275. All the results are when the fuel scaled average is 1 t/d. (Fig. 13, Fig. 14, Fig. 15., Fig. 16.).

The results recorded minor changes in the NPC and COE if the biomass fuel scaled average is maximum (1.5 t/d). The results are shown in Fig. 17 and 18 for NPC and COE respectively.

The sensitivity analysis demonstrates the impact of varying multiple variables to identify their influence on NPC and COE. Decision or sensitivity variable values can be selected based on specific requirements. Moreover, the combination of evaluated sensitivity factors defines the different sensitivity cases.

4.3.3. Uncertainty consideration

To account for uncertainties in microgrid analysis, stochastic models that incorporate the variability of wind and load within the microgrid have been developed. Given the limitations of HOMER in directly addressing these uncertainties, MATLAB was employed to construct stochastic models and generate the corresponding wind speed and load scenarios. These generated data were then integrated into HOMER to validate the performance of the designed microgrid. The generation of wind speed and load scenarios can be described as the following subsections. Additionally, the analysis of the microgrid has been conducted by incorporating these stochastic scenarios to assess its performance under varying conditions.

• Wind Speed Scenarios generation

The energy output of Wind Turbine (WT) units is influenced by wind speed, which is an inherently uncertain environmental parameter. The probability distribution of wind speed can be represented using the Weibull distribution, as described in the following equation [44].

$$f(v|c_{wind}, \lambda_{wind}) = \frac{c_{wind}}{\lambda_{wind}} \left(\frac{v}{\lambda_{wind}} \right)^{c_{wind}-1} e^{-\left(\frac{v}{\lambda_{wind}} \right)^{c_{wind}}} \quad (20)$$

The parameters of the Weibull distribution, which are critical for accurately modeling wind speed behavior, can be estimated using the mathematical formulations provided in the following equations.

$$c_{wind} = \left(\frac{\sigma_{wind}}{\mu_{wind}} \right)^{-1.086} \quad (21)$$

$$\lambda_{wind} = \frac{\mu_{wind}}{\Gamma\left(1 + \frac{1}{c_{wind}}\right)} \quad (22)$$

These equations facilitate the determination of the shape parameter λ_{wind} and scale parameter c_{wind} , which define the distribution's characteristics. Accurate estimation of these parameters is essential for capturing the variability and probabilistic nature of wind speed, as highlighted in studies [44].

The mathematical representation of the Weibull Cumulative Distribution Function (CDF), which describes the probability of wind speed being less than or equal to a given value, is detailed as [44];

$$F(v|c_{wind}, \lambda_{wind}) = 1 - e^{-\left(\frac{v}{\lambda_{wind}} \right)^{c_{wind}}} \quad (23)$$

The inverse of the Weibull Cumulative Distribution Function (CDF) is employed to simulate hourly wind speed by generating random values that align with the specified probability distribution. This approach, as illustrated in equation (14), ensures that the simulated data accurately reflects the stochastic nature of wind speed.

$$v = F^{-1}(u|c_{wind}, \lambda_{wind}) = -\lambda_{wind} \ln(1 - u)^{\frac{1}{c_{wind}}} = -\lambda_{wind} \ln(u)^{\frac{1}{c_{wind}}} \quad (24)$$

Fig. 19 shows the histograms and time-series plots to compare wind speeds generated using the Weibull distribution with real data, showing that the simulated distributions closely follow the observed trends. This highlights the effectiveness of the Weibull distribution in modeling wind speeds, though some discrepancies may arise from real-world variations not captured by the parameters estimated through Maximum Likelihood Estimation (MLE). The consistency in variability between simulated and real data supports the model's reliability while incorporating additional uncertainty factors could enhance its accuracy [44].

• Electrical Load Models considering the uncertainty

The electrical load is modeled as random variables following a normal distribution, consistent with observed patterns in real-world data. The mean and standard deviation of the load demand are calculated directly from historical data, ensuring that the simulated values reflect the typical characteristics and variability of actual demand. Using

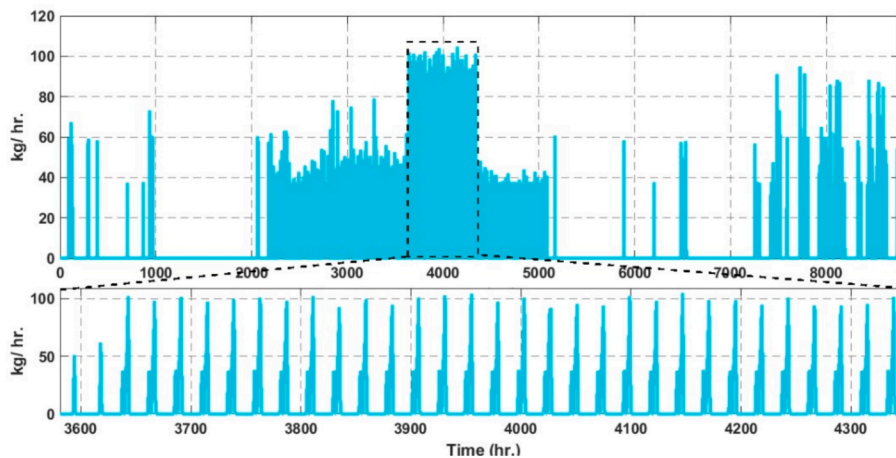


Fig. 12. Yearly Biomass Consumption.

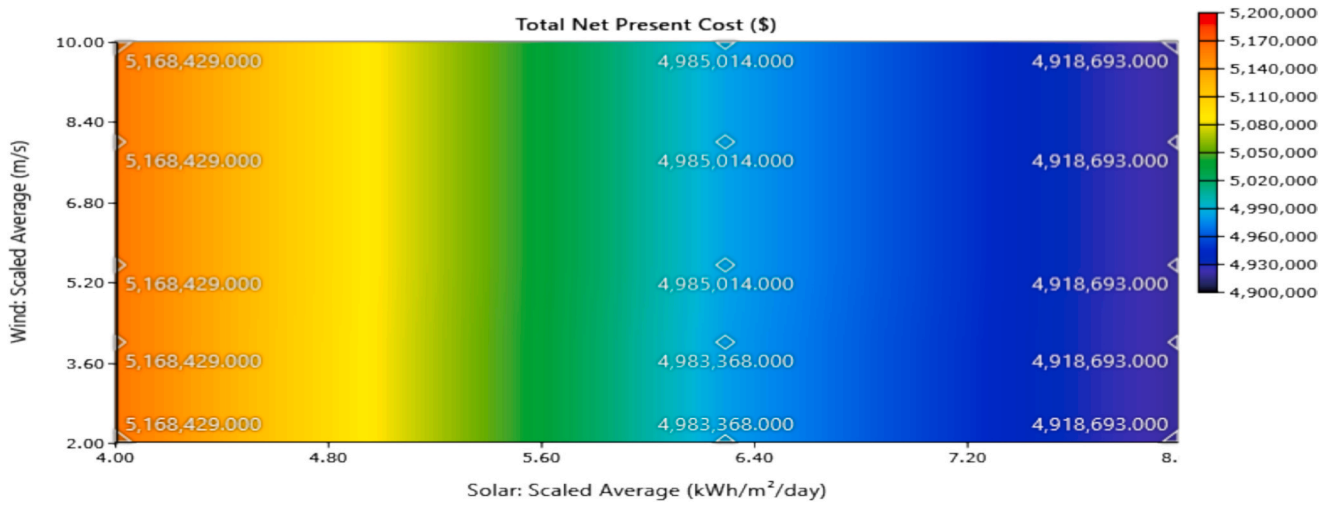


Fig. 13. NPC values considering the fluctuations of the wind speed rate and solar radiation rate under average temperature of 28 °C.

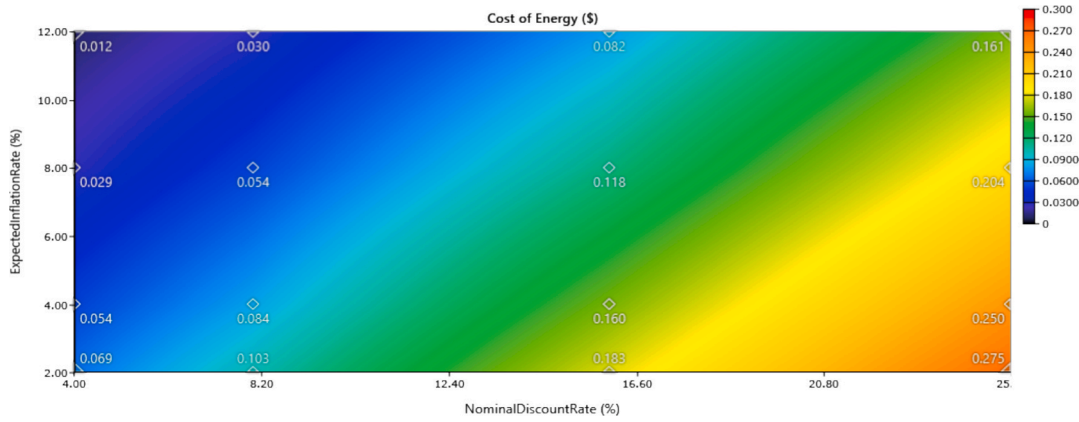


Fig. 14. COE values considering the fluctuations of the wind speed rate and solar radiation rate under the average temperature of 28 °C.

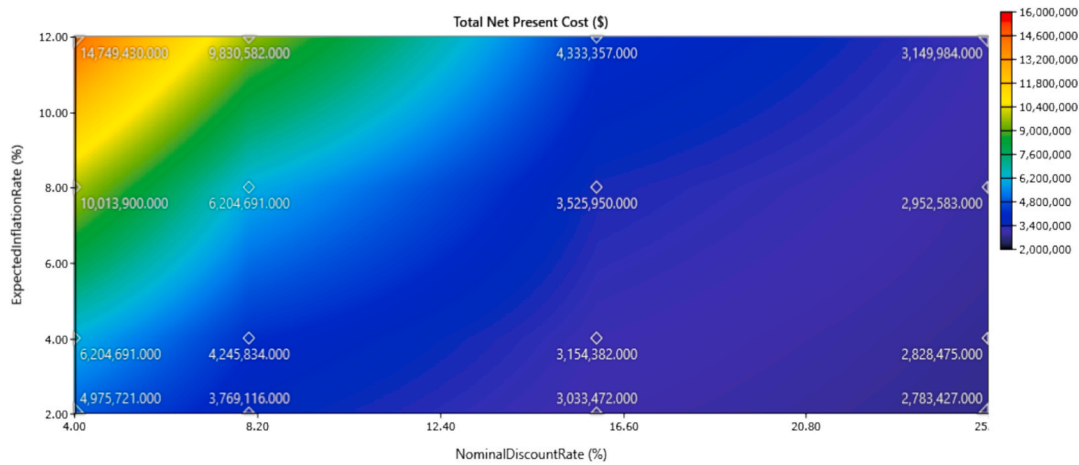


Fig. 15. NPC values considering the fluctuations of the nominal discount rate and inflation (numbers inside the surface plot) under a biomass fuel rate of 1 t/d.

Monte Carlo simulation, synthetic datasets for electrical load demand are generated for an entire year (8760 data points), providing a comprehensive representation of hourly load behavior. So, the load demand is modeled using normal distribution as follows [45].

$$F_{load}(L_d) = \frac{1}{\sqrt{2\pi}\sigma_{L_d}} \exp\left(-\frac{(L_d - L_{dmean})^2}{2\sigma_{L_d}^2}\right) \quad (25)$$

Fig. 20 shows the histograms to compare the probability distributions of electrical loads generated using a normal distribution with the real electrical load data. The figure also presents time-series plots for a

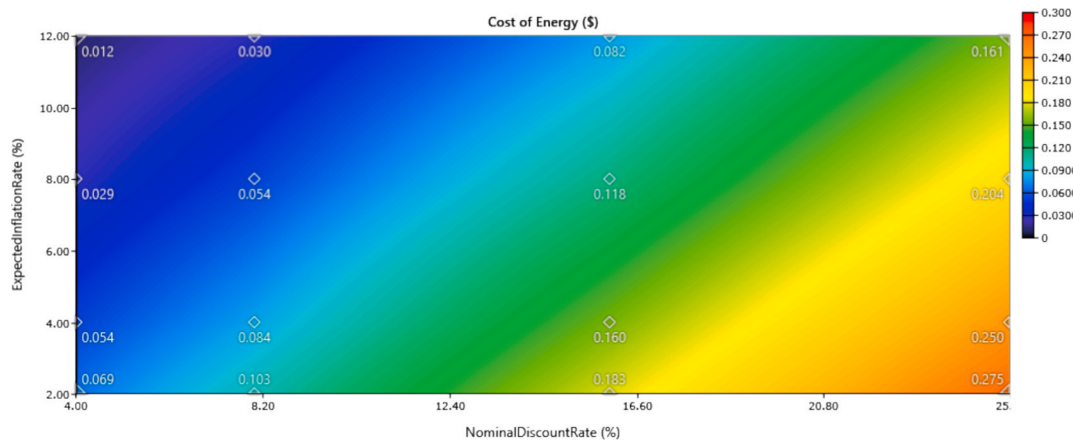


Fig. 16. COE values considering the fluctuations of the nominal discount rate and inflation (numbers inside the surface plot) under a biomass fuel rate of 1 t/d.

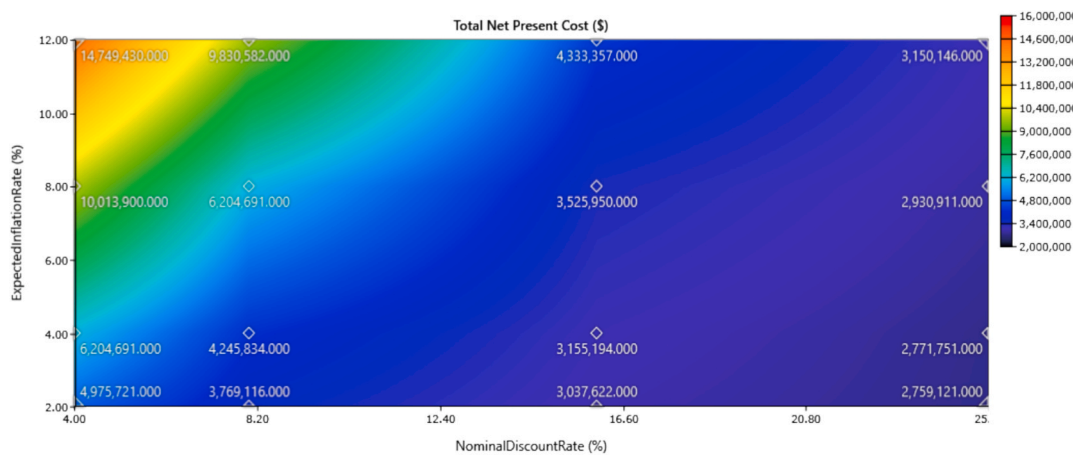


Fig. 17. NPC values considering the fluctuations of the nominal discount rate and inflation (numbers inside the surface plot) under a biomass fuel rate of 1.5 t/d.

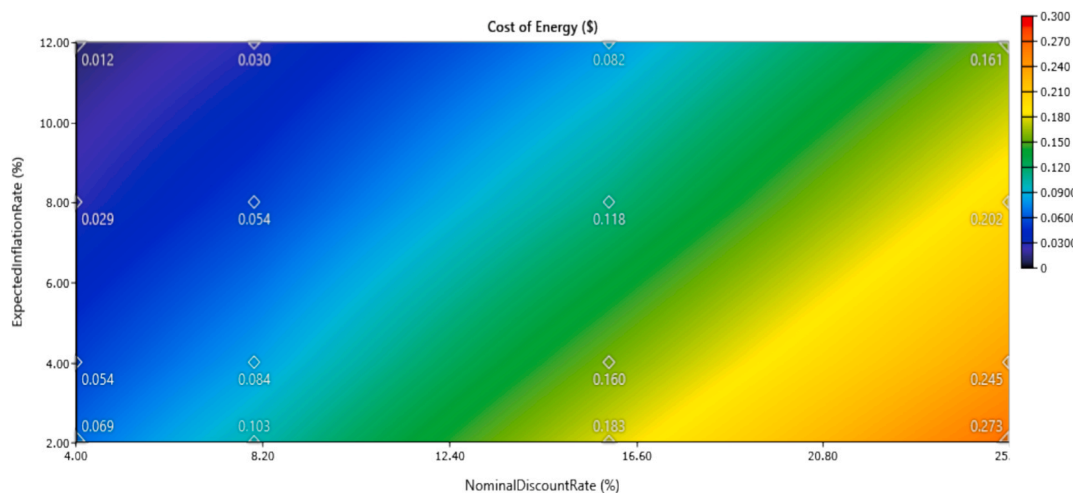


Fig. 18. COE values considering the fluctuations of the nominal discount rate and inflation (numbers inside the surface plot) under a biomass fuel rate of 1.5 t/d.

visual comparison of the simulated and real data. While the generated loads capture the mean and standard deviation of the real data [45]. Table 10 compares nine scenarios, including a base scenario, by presenting the results of the Net Present Cost (NPC) and Cost of Energy (COE). The results also demonstrate each case's excess energy, unmet load, and LPSP probability. The table also shows each component rating,

production and operating hours. The system is reliable in all cases while the highest and lowest values of NPC are in cases Wind2-Load1 (3,994,892 \$) and Wind1-Load1 (3,346,282 \$) respectively.

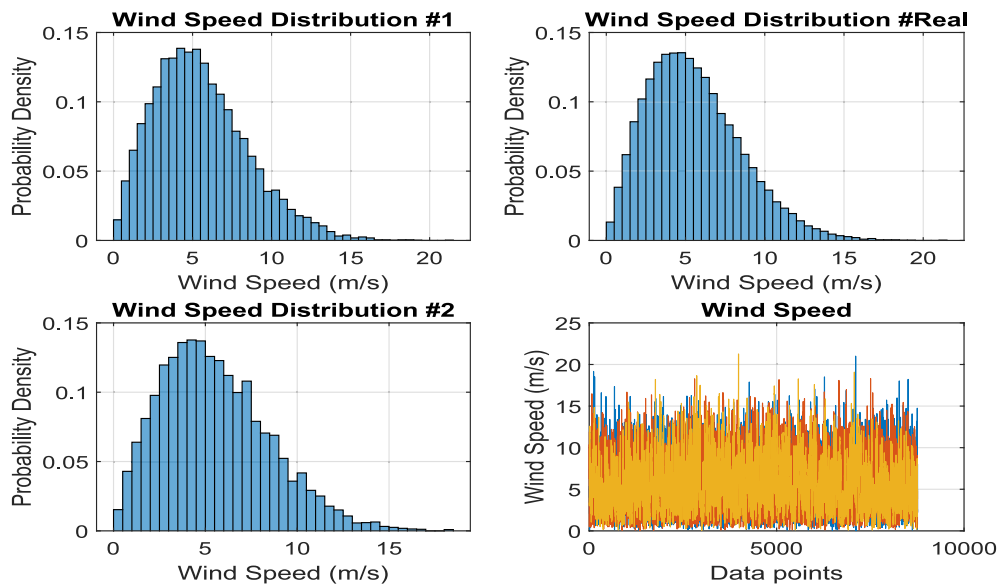


Fig. 19. Histograms and time-series plots comparing Weibull-generated and real wind speed data.

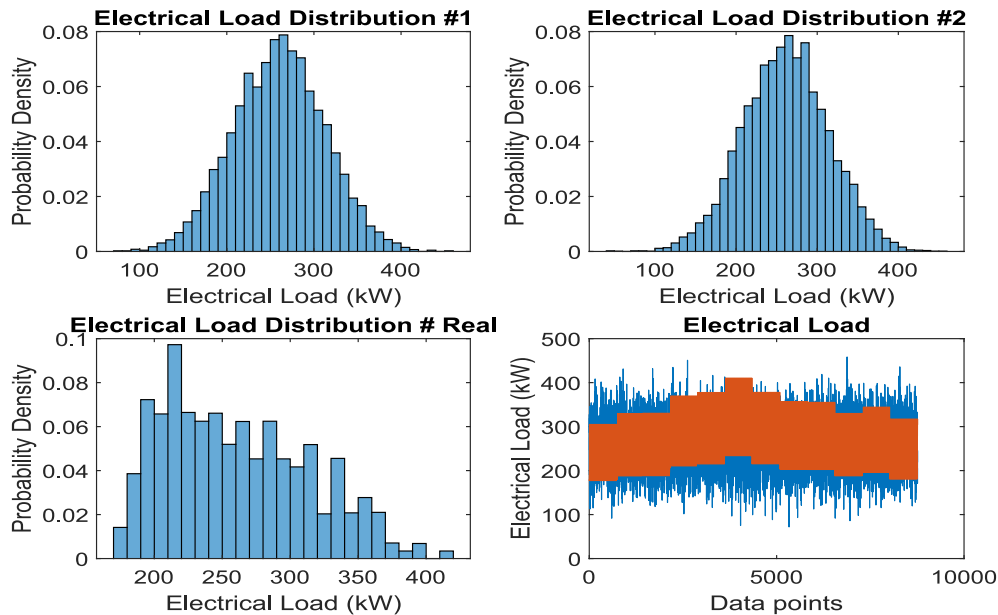


Fig. 20. Comparison of histograms and time-series plots for electrical loads generated by a normal distribution and real data.

5. Conclusions

This paper investigates the optimized design of a HRES with a backup biomass generator in the remote area of Abu-Minqar in Egypt. The HOMER Pro software is used to perform comprehensive optimization, resulting in remarkably favorable findings. The results demonstrate that the proposed system design efficiently utilizes renewable energy resources (photovoltaic (PV) panels, WT, batteries, and system converter), as well as the backup biomass generator. A thorough load analysis and cautious system sizing ensure that the system has sufficient capacity and can successfully meet all electrical loads in Abu-Minqar. The applied design significantly improved the system performance by precisely modeling renewable resource potential and using optimization techniques to minimize excess energy while maintaining a reliable and continuous supply of electricity. The obtained exceptional results demonstrate the viability and efficiency of HRESs in isolated rural areas. Combining renewable energy sources with a backup biomass generator

provides a sustainable way to meet electricity demand, diminish reliance on fossil fuels as well as mitigate environmental impact. The designed energy system achieves the lowermost NPC and LCOE associated with other studies in the same area for supplying the same load profile. This is due to the use of the biomass generator as a backup source for only 505 h/year. The proposed system can be utilized to decrease the emission of harmful gases that are contributing to global warming. Environmental aspects are taken into account when modeling a biomass-solar off-grid hybrid system. Overall, the presented framework contributes to the growing body of knowledge of HRESs besides provides policymakers, researchers, and industry practitioners with valuable information on the implementation of such systems in rural areas. In future work, other configurations including smart irrigation systems, EVs and green hydrogen have been planned to be studied and analyzed based on optimization techniques and AI-based methods. In the future, techno-economic environmental analysis of a microgrid with several cases using various algorithms must be conducted to enhance system

Table 10

Different scenarios and cases due to changes in the load and wind speed curves.

Parameter	wind-load	Wind-Load1	Wind-Load2	Wind1-Load	Wind1-Load1	Wind1-Load2	Wind2-Load	Wind2-Load1	Wind2-Load2
LPSP	0	0.0005	0	0	0	0	0	0	0
Capacity Shortage KWh/year	0	11.3	0	0	0	0	0	0	0
unmet loads kWh/year	0	0	0	0	0	0	0	0	0
Excess Energy kWh/year	804,156	1,181,955	879,082	800,112	1,363,304	873,406	774,880	1,363,304	875,100
Excess Energy %	23	30.9	25	23	34.1	24.9	22.4	34.1	24.9
Production(Kwh/Year)	3,486,265	3,822,539	3,515,089	3,481,202	3,994,892	3,508,857	3,454,569	3,626,280	3,510,249
NPC (\$)	3,483,519	3,357,720	3,490,807	3,357,501	3,346,282	3,486,574	3,474,901	3,994,892	3,483,939
COE (\$/kWh)	0.118	0.114	0.119	0.118	0.113	0.119	0.118	0.113	0.119
BIOMASS Generator									
Rating (kW)	460	430	460	460	430	460	460	430	460
operating hours	505	1121	871	485	1144	862	524	1144	854
no. of starts	214	782	584	209	806	575	219	806	572
Electric Production (KWh/year)	93,808	280,739	216,403	89,769	295,998	211,196	96,477	295,998	210,939
Photovoltaic									
Rating (KW)	1749	1828	1700	1749	1911	1700	1731	1911	1700
Electric Production (kWh/year)	3,318,818	3,468,161	3,225,046	3,318,818	3,626,280	3,225,046	3,283,828	3,626,280	3,225,046
operating hours (h/year)	4,378	4,378	4,378	4378	4,378	4,378	4,378	4,378	4,378
Wind Turbine									
Rating (kW)	1 x 30	1 x 30	1 x 30	1 x 30	1 x 30	1 x 30	1 x 30	1 x 30	1 x 30
Electric Production (kWh/year)	73,639	73,639	73,639	72,615	72,615	72,615	74,264	72,615	74,264
operating hours (h/year)	7888	7888	7888	7816	7816	7816	7,867	7816	7867
Batteries									
Qty	1560	1,320	1440	1560	1,320	1440	1520	1320	1440
Energy out (kWh/year)	1,159,177	1,034,116	1,085,206	1,153,996	1,001,180	1,081,886	1,149,051	1,001,180	1,080,683
operating hours (Autonomy)	24.8	21	23.1	24.8	21	23.1	24.2	21	23.1

performance and identify the most effective optimization method.

CRedit authorship contribution statement

Mohamed M. Torad: Conceptualization, Data curation, Formal analysis, Methodology, Software, Writing – original draft. **Sayed H.A. Elbanna:** Conceptualization, Writing – review & editing, Supervision. **Mahmoud A. El-Dabah:** Conceptualization, Writing – review & editing, Supervision. **Ahmed A. Zaki Diab:** Conceptualization, Writing – review & editing, Supervision. All authors have read and agreed to the published version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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